UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

© Hak Cipta Universiti Teknologi MARA

PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

Complete the following taxonomy of parallel processor architecture with appropriate example.
Single Instruction Multiple
Data Stream
ii
Processor Organization
Multiple Instruction Multiple
Data Stream
v

Source: CSC429 (5).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Computer generations and history

The X machine was the first automatic electronic digital computer that uses binary system. It has 30 arithmetic/logic unit that can add and subtract, a rotating drum memory that can held thirty binary numbers of fifty digits each and input/output components. Figure 1 shows the architecture of X machine.
bose-iO to
drum timing contot
drum
base -2
contacts
brushes
boost carry-
dYVhf
brushes
motor
switch
r box
gut
pwer suppty
regulator
crd
(7 duel triodes)
circuit
Figure 1: Machine X
a) What is X machine?
b) Name the inventor of X machine.
c) Based on Figure 1, identify the input and output components for X machine.

Source: CSC429 (7).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Data representation and number systems

Calculate -53 - 31 using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

Answer: left -53 -> 1 0110101; right 31 -> 0 0011111; -53 - 31 = -84 -> sign-magnitude 1 1010100

• *1 mark for converting both operands to sign-magnitude.*

© Hak Cipta Universiti Teknologi MARA

- *1 mark for applying the sign rule for subtraction.*
- *1 mark for the correct magnitude operation.*
- *1 mark for the correct sign bit.*
- *1 mark for the final decimal and byte representation.*

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

Figure 1 illustrates the icon of one of the interfaces to the external devices.
ffi /^
>
Figure 1: Interface to the external devices
a) Identify the type of external device interface based on the icon in Figure 1.
b) The external devices interface in (a) can have a "hot connection". Define hot connection.
c) Discuss the THREE (3) input-output techniques available in a computer system and elaborate on the Central Processing Unit's involvement in each technique.

Source: CSC429 (17).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Memory hierarchy and cache

a) Rank the following memory types based on price ranging from the cheapest to the most expensive.
Cache, tape back-up, register, hard drive
b) Discuss ONE (1) benefit of cache memory.
c) Differentiate between write through and write back mechanisms in order to store data in cache.

Source: CSC429 (8).pdf, page(s) 4, 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the whole operation of the machine cycle and its respective responsible computer components.
b) Elaborate on what will happen in the machine cycle if the main memory is corrupted.

Source: CSC429 (21).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source

© Hak Cipta Universiti Teknologi MARA

paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

Input output technique can be segregated into programmed, interrupt-driven and direct memory access (DMA).
a) What is DMA?
b) How can the DMA technique improve the overall computer performance?
c) Discuss TWO (2) major drawbacks of programmed and interrupt-driven input output technique that inspire hardware developer to introduce DMA.
d) Conventional access is depicted in Figure 2. Redraw Figure 2 to illustrate DMA architecture.
Primary Memory
CPU
Controller
Device
Figure 2: Conventional access architecture

Source: CSC429 (2).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a)
List TWO (2) buses types that are available in the computer system.
b)
Differentiate TWO (2) characteristics of system interconnection
and CPU
interconnection.
c)
Explain the different types of data transfer timing.

Source: CSC429 (15).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.

a) Convert -31.5 to IEEE 754 single precision. Give the answer in hexadecimal.

Answer: -31.510 = 0xC1FC0000 (sign=1, exponent=10000011, fraction=11111000000000000000000)

- *1 mark for sign, exponent and mantissa.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the 32-bit binary grouping.*
- *1 mark for the hexadecimal result.*

b) Convert the IEEE 754 single-precision value 0xC1360000 to decimal.

Answer: 0xC1360000 = -11.37510

- *1 mark for the sign bit.*
- *1 mark for the exponent field.*
- *1 mark for the fraction field.*
- *1 mark for applying the bias and significand.*
- *1 mark for the final decimal value.*

© Hak Cipta Universiti Teknologi MARA

PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 1 and Table 1.
z=A2-B*C+jl_
D+5
Figure 1
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: Machine instruction
Three-address
One-address
Zero-address
ADD (p,q,r)
LOAD p
PUSH p
suB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

Source: CSC429 (20).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

© Hak Cipta Universiti Teknologi MARA